

STM32 Training Manual

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Board: *STM32F407G-DISC1*

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1. What is STM32?

STM32 is a family of **32-bit ARM Cortex-M microcontrollers** developed by **STMicroelectronics**. STM32 MCUs are widely used in embedded systems due to:

- High processing speed
- Low power consumption
- Rich peripheral set (GPIO, UART, SPI, I2C, ADC, Timers, USB, etc.)
- Excellent real-time performance

The **STM32F4 Series**, including the STM32F407, uses a **Cortex-M4 core @ 168 MHz**, making it ideal for learning and advanced applications.

2. STM32F407G-DISC1 Board Overview

The STM32F407 Discovery board provides everything needed to start embedded development.

Key Hardware Components

- **STM32F407VGT6 MCU** (ARM Cortex-M4, 168 MHz, FPU)
- **4 User LEDs**
 - LD3 (Orange) → PD13
 - LD4 (Green) → PD12
 - LD5 (Red) → PD14
 - LD6 (Blue) → PD15
- **User Button (B1)**
- **ST-LINK/V2** On-board Programmer/Debugger
- **On-board peripherals:**
 - MEMS accelerometer
 - Microphone
 - Audio DAC
 - USB OTG Full-Speed
 - I2S, SPI, I2C headers

LED Pin Summary

LED	Color	GPIO Pin
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LD4	Green	PD12
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LD3	Orange	PD13
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LD5	Red	PD14
-----	-----	------

LD6	Blue	PD15
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These LEDs are perfect for your first GPIO project.

3. Installing STM32CubeIDE

STM32CubeIDE is ST's official integrated development environment. It includes:

- STM32CubeMX (pin configuration + code generator)
- C/C++ editor
- Compiler
- Debugger
- ST-Link support

Steps to Install

1. Visit: <https://www.st.com/stm32cubeide>
2. Download for your OS (Windows/Linux/macOS).
3. Run the installer.
4. Install **ST-Link USB drivers** if prompted.
5. Launch STM32CubeIDE.

Additional Tools

- **Serial Monitor:** Tera Term / PuTTY
 - **USB cables:**
 - Mini-USB → ST-LINK
 - Micro-USB → USB OTG FS (for CDC output)
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4. Project 1 — LED Blinking on STM32F407G-DISC1

This introductory project demonstrates basic GPIO control by blinking all four user LEDs on the STM32F407 Discovery board. The LEDs connected to pins **PD12, PD13, PD14, and PD15** will blink five times together, after which the **green LED (PD12)** will continue blinking indefinitely.

4.1 Hardware Required

- **STM32F407G-DISC1** development board
- **Mini-USB cable** (for ST-LINK programming + power)

- No additional components required
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4.2 Creating a New STM32 Project

1. Open **STM32CubeIDE**
2. Navigate to **File** → **New** → **STM32 Project**
3. In the Board Selector, search for **STM32F407G-DISC1**
4. Select the board and click **Next**
5. Enter the project name: **LED_Blink_Project**
6. Click **Finish**

STM32CubeMX configuration view will open automatically.

4.3 Configuring GPIO Pins

Configure the four on-board LEDs:

1. In the Pinout view, select the following pins:
 - **PD12**
 - **PD13**
 - **PD14**
 - **PD15**
2. Set each pin to **GPIO_Output**

CubeMX will handle peripheral initialization and GPIO clock configuration automatically.

4.4 Generating Code

Click the **Generate Code** (gear icon).

The IDE will produce all required source files, including **main.c**, which contains the application entry point.

4.5 Writing the LED Blink Code

Open **main.c** and locate the following loop:

```
while (1)
{
    /* USER CODE BEGIN 3 */

    /* USER CODE END 3 */
}
```

Replace the section between **USER CODE BEGIN 3** and **USER CODE END 3** with the following code:

```
// Blink all LEDs 5 times
for (int i = 0; i < 5; i++)
{
    HAL_GPIO_WritePin(GPIOD, GPIO_PIN_12, GPIO_PIN_SET);

    HAL_GPIO_WritePin(GPIOD, GPIO_PIN_13, GPIO_PIN_SET);

    HAL_GPIO_WritePin(GPIOD, GPIO_PIN_14, GPIO_PIN_SET);

    HAL_GPIO_WritePin(GPIOD, GPIO_PIN_15, GPIO_PIN_SET);

    HAL_Delay(500);
}
```

```
    HAL_GPIO_WritePin(GPIOD, GPIO_PIN_12, GPIO_PIN_RESET);

    HAL_GPIO_WritePin(GPIOD, GPIO_PIN_13, GPIO_PIN_RESET);

    HAL_GPIO_WritePin(GPIOD, GPIO_PIN_14, GPIO_PIN_RESET);

    HAL_GPIO_WritePin(GPIOD, GPIO_PIN_15, GPIO_PIN_RESET);

    HAL_Delay(500);

}

// After 5 blinks, blink Green LED forever
while (1)
{
    HAL_GPIO_TogglePin(GPIOD, GPIO_PIN_12);

    HAL_Delay(300);
}
```

What the code does

- Turns all four LEDs ON and OFF together, five times
- After five cycles, the program enters an infinite loop
- The **green LED (PD12)** toggles indefinitely as a heartbeat indicator

4.6 Building & Flashing the Program

1. Click the **Build** button (hammer icon).
2. Connect the board via **Mini-USB** to the ST-LINK port.

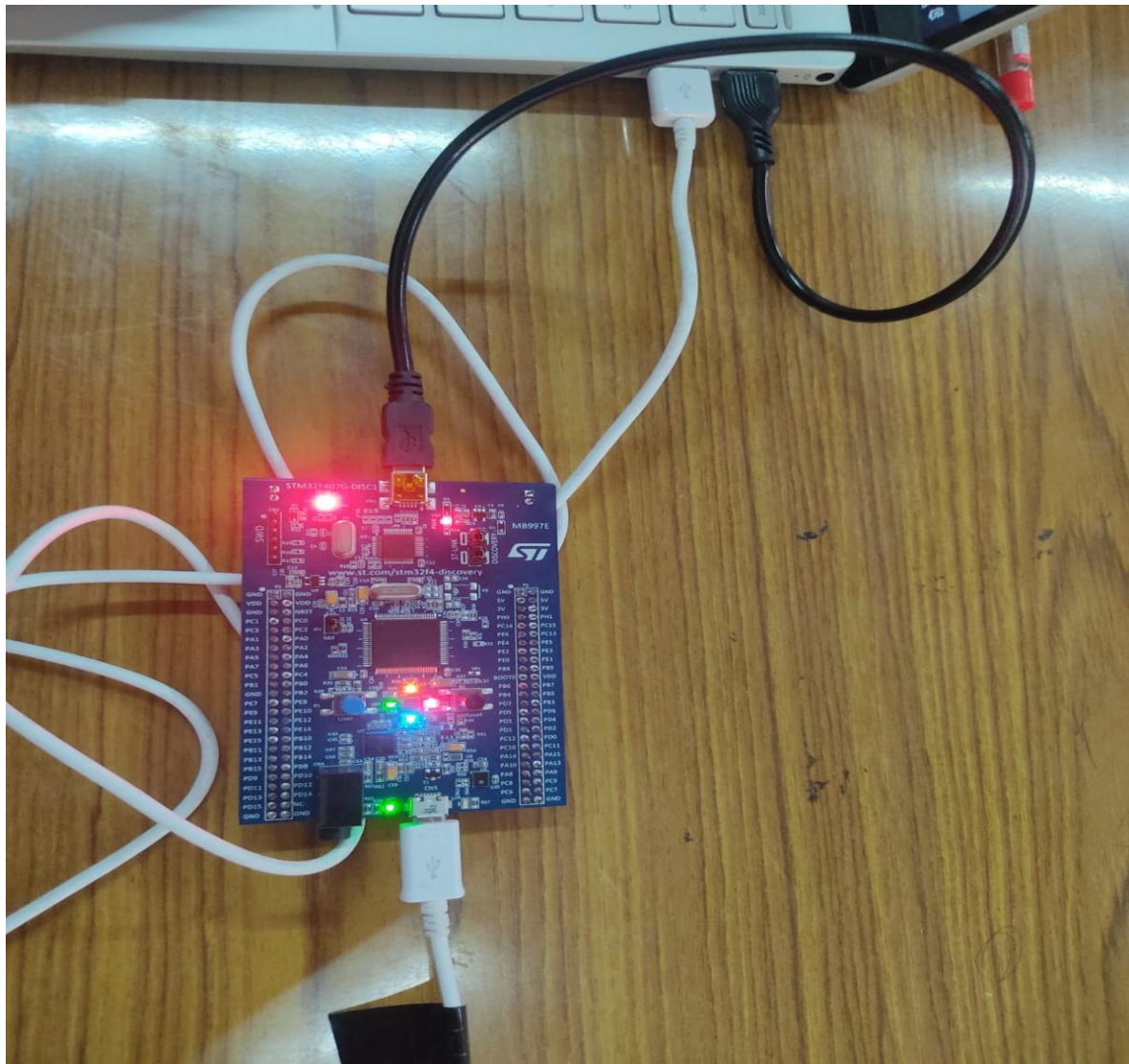
3. Click **Run** → **Debug** (or *Run As* → *STM32 MCU Application*)

The firmware will be programmed into the MCU and will start running automatically.

Expected Behavior

- All four LEDs (PD12–PD15) blink **simultaneously five times**
 - After that, **only the green LED (PD12)** will continue blinking forever at 300 ms intervals
-

Output



5. Project 2 — USB CDC “Hello from STM32 via USB!”

This project teaches USB communication by making the board appear as a **Virtual COM Port** and send text to a computer.

5.1 Hardware Required

- STM32F407G-DISC1
 - **Mini-USB** cable → ST-LINK
 - **Micro-USB** cable → USB OTG FS
 - **Tera Term** or **PuTTY**
-

5.2 Understanding the USB Ports

USB Port	Function
Mini-USB (ST-LINK)	Power + Flashing + Debugging
Micro-USB (USB OTG FS)	USB CDC (serial data output)

Both cables must be connected.

5.3 Creating a New Project

1. File → New → STM32 Project
 2. Select **STM32F407G-DISC1**
 3. Name the project: `USB_CDC>HelloWorld`
 4. Finish
-

5.4 Configuring USB Device (CDC)

Step 1 — Enable USB OTG FS

1. Go to **Connectivity** → **USB_OTG_FS**
2. Set:
Mode = Device_Only

CubeMX assigns:

- PA11 → USB_DM
 - PA12 → USB_DP
-

Step 2 — Enable USB Device Middleware

1. Click **Middleware** → **USB_DEVICE**
2. Set:
Class For FS = Communication Device Class (CDC)

This generates:

- `usbd_cdc_if.c`

- `usbd_cdc_if.h`
 - USB device stack
-

5.5 Generate Code

Click **Generate Code**.

5.6 Writing the USB Hello World Code

Add Includes

At the top of `main.c`, inside **USER CODE BEGIN Includes**:

```
#include "usbd_cdc_if.h"

#include <string.h>
```

Write USB Message Logic

Place inside the **while(1)** loop:

```
char msg[] = "Hello from STM32 via USB!\r\n";

// Wait for USB configuration
while (hUsbDeviceFS.dev_state != USB_STATE_CONFIGURED) {
    HAL_Delay(10);
}

// Send message 10 times
```

```
for (int i = 0; i < 10; ++i) {  
  
    uint8_t res;  
  
    uint8_t tries = 0;  
  
    do {  
  
        res = CDC_Transmit_FS((uint8_t *)msg, (uint16_t)strlen(msg));  
  
        if (res == USBD_OK) break;  
  
  
        if (res == USBD_BUSY) {  
  
            HAL_Delay(5);  
  
            tries++;  
  
        } else {  
  
            break;  
  
        }  
  
    } while (tries < 10);  
  
  
    HAL_Delay(1000);  
  
}
```

// After sending messages, blink LED forever

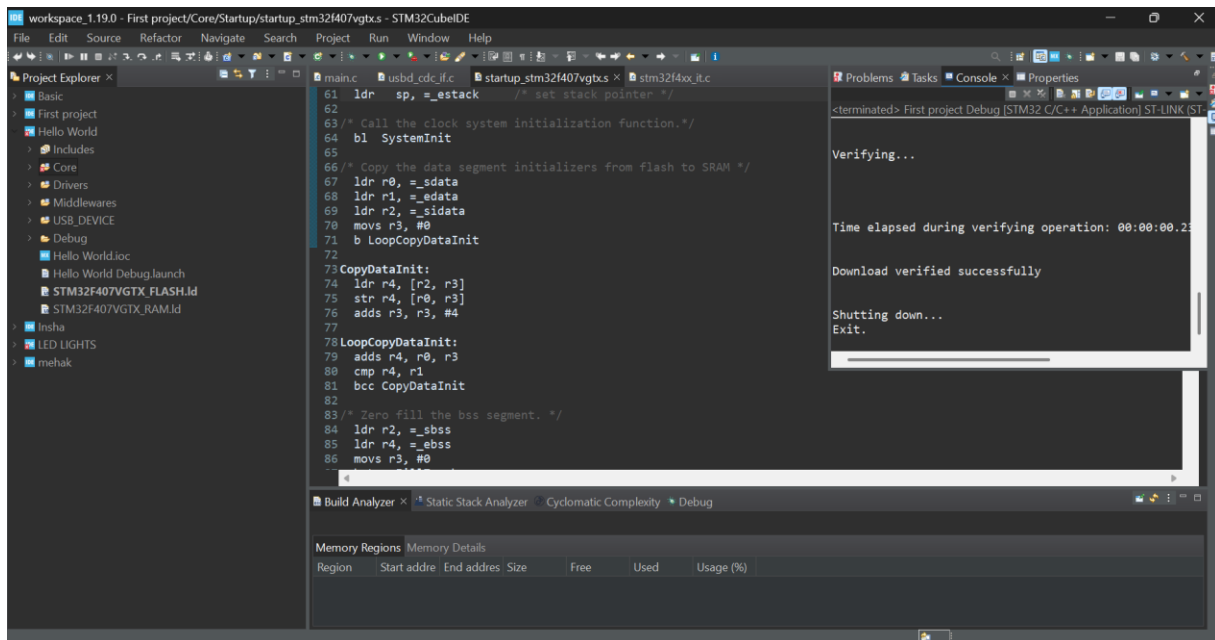
```
while (1) {  
  
    HAL_GPIO_TogglePin(GPIOD, LD4_Pin);  
  
    HAL_Delay(300);  
  
}
```

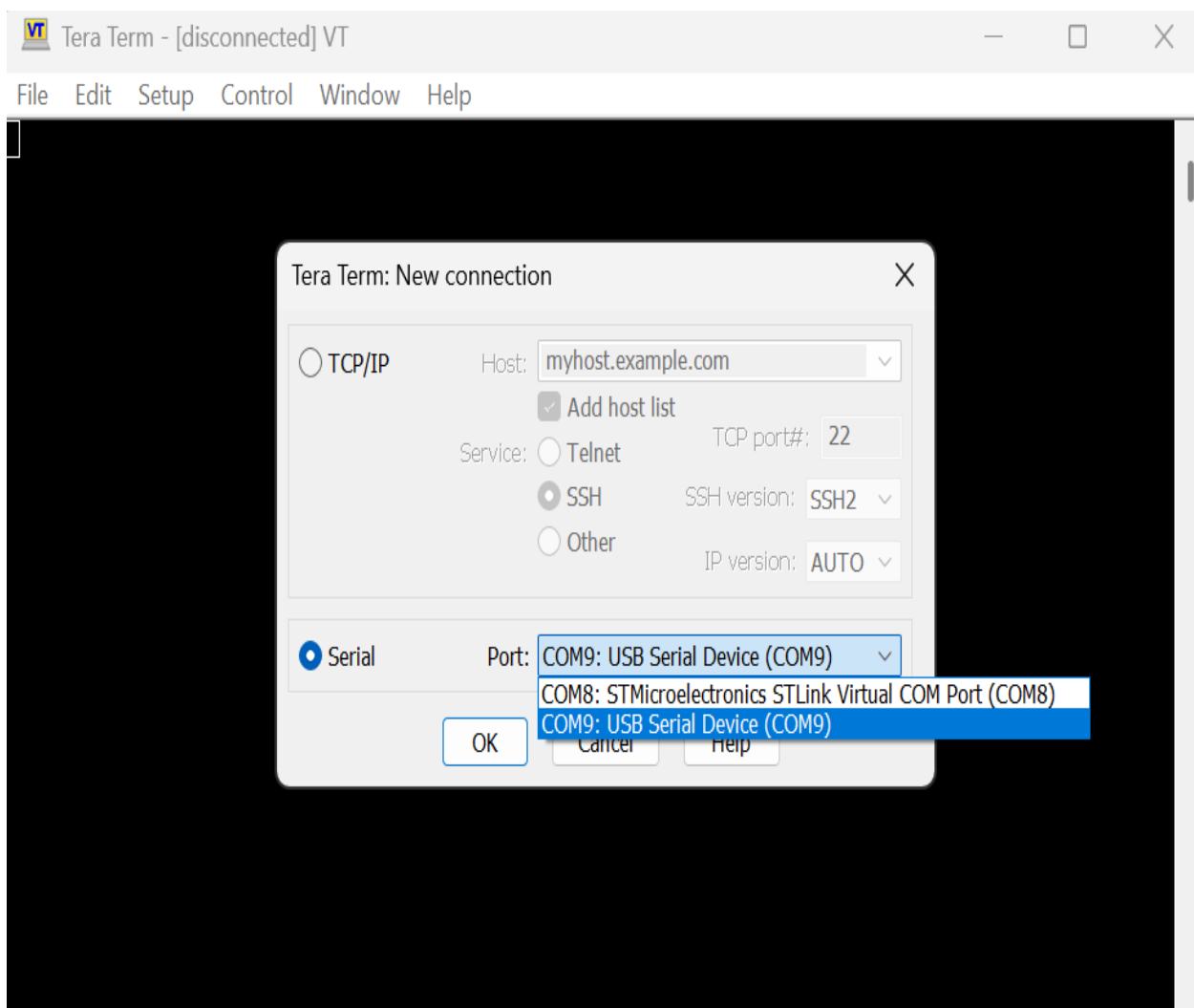
}

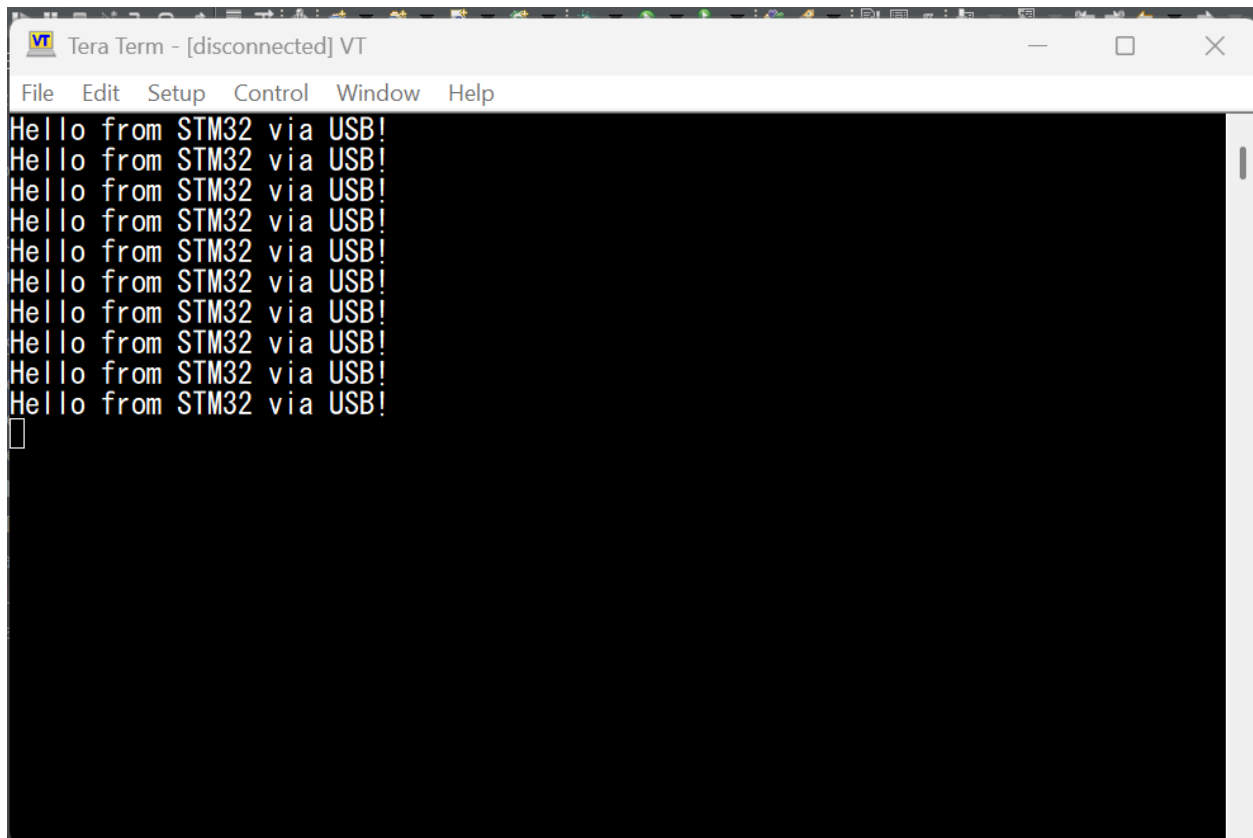
5.7 Build, Flash, and Run

1. Build the project
2. Flash using **Run** → **Debug**
3. Connect the **Micro-USB** cable
4. Open **Tera Term** → Select:
USB Serial Device (COMx)

Output







The image shows a screenshot of a Tera Term terminal window. The title bar reads 'Tera Term - [disconnected] VT'. The menu bar includes 'File', 'Edit', 'Setup', 'Control', 'Window', and 'Help'. The terminal area has a black background with white text. It displays ten lines of the message 'Hello from STM32 via USB!'. A white cursor is positioned at the start of the eleventh line, which is currently empty.

Then LD4 blinks forever.

6. Troubleshooting Guide

USB CDC Not Working

- ✓ Connect Micro-USB cable to **USB OTG FS**
- ✓ Check correct COM port
- ✓ Re-open terminal after flashing
- ✓ Ensure CubeMX config = CDC + Device_Only

CDC_Transmit_FS returns BUSY

- ✓ Terminal must be open
- ✓ Increase retry delay
- ✓ Ensure USB Device is configured

LEDs Not Blinking

- ✓ Pins PD12-PD15 set as GPIO_Output
- ✓ HAL_Delay works
- ✓ ST-Link drivers installed